

ASSIGNMENT 3- AI-Powered Fleet Performance & Delivery Efficiency Dashboard(POWER BI)

1

Assignment Overview:

A logistics company wants to analyze its fleet performance in terms of on-time deliveries, fuel efficiency, and cost per mile.

Dataset : The given dataset contain following fields to perform the given tasks,
Dataset Fields:

- Trip_data: Trip ID, Vehicle ID, Driver ID, Origin, Destination, Distance (km), Fuel Consumed (liters), Delivery Status (On-Time/Late), Delivery Date.
- Vehicle Master: Vehicle ID, Vehicle Type, Capacity, Maintenance Cost.

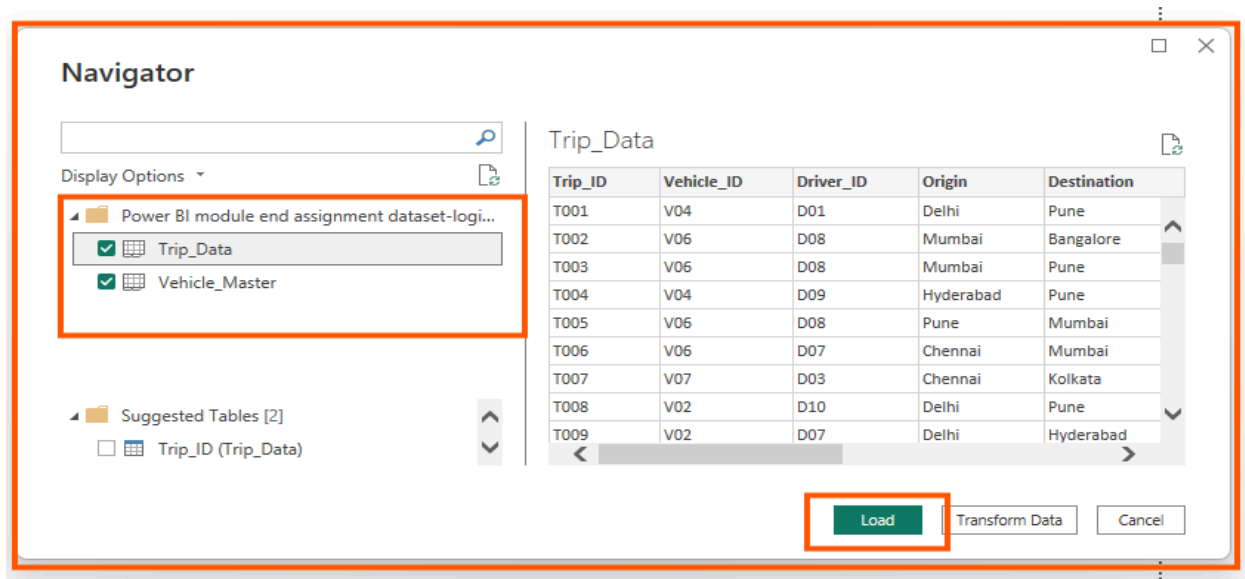
Task 1:

1. Data Cleaning & Modeling:

- Fix missing fuel consumption values (use mean imputation).
- Create a relationship using Vehicle_ID between Trip_Data and the Vehicle Master table.

Master table.

At first I imported the trip_data into Power Query editor and then also imported the Vehicle_master table and loaded it in the power query editor.



And then I checked for any missing values in the Trip_data table and also for the Vehicle_data table using sorting by right clicking in the Fuel_consumed_L column from the Trip_data table.

The given dataset of Trip_data table and Vehicle_Master tables are

Query Settings

Table.ReplaceValue(#"Kept First Rows",null,91.88,Replacer.ReplaceValue,{"Fuel_Consumed_L"})

	Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date
1	hi	Pune	117	108.42	On-Time	
2	mbai	Bangalore	172	161.33	On-Time	
3	mbai	Pune	145	154.7	On-Time	
4	derabad	Pune	38	26.6	On-Time	
5	he	Mumbai	39	33.2	On-Time	
6	ennai	Mumbai	127	85.04	late	
7	ennai	Kolkata	75	58.08	On-Time	
8	hi	Pune	7	5.24	On-Time	
9	hi	Hyderabad	18	16.22	On-Time	
10	galore	Hyderabad	187	105.21	late	
11	kata	Hyderabad	41	31.17	On-Time	
12	kata	Delhi	75	51.77	On-Time	
13	kata	Chennai	157	188.52	late	
14	derabad	Bangalore	152	104.51	On-Time	
15	kata	Mumbai	195	179.88	On-Time	
16	galore	Mumbai	85	91.88	late	
17	he	Kolkata	126	102.91	On-Time	
18	he	Delhi	156	107.23	On-Time	
19	derabad	Pune	179	155.52	On-Time	
20	mbai	Bangalore	164	148.87	On-Time	
21	hi	Mumbai	44	31.98	late	
22	derabad	Delhi	14	11.44	On-Time	
23	ennai	Delhi	177	107.55	On-Time	

Query Settings

PROPERTIES

Name

Trip_Data

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Kept First Rows

FuelConsumed Replaced Value

Cleaned using Power Query in Microsoft Excel. I calculated the average fuel consumption and replaced the null value with the mean value of 91.88. as it is fuel related decimal points so i have kept 2 decimal points for accurate calculations of fuel Price.

The relationship using Vehicle_ID between Trip_Data and the Vehicle Master table is created using Vehicle_ID.

New relationship

Select tables and columns that are related.

From table

Vehicle_Master

Capacity_kg	Maintenance_...	Vehicle_ID	Vehicle_Type
8424	5633	V01	Van
1970	6776	V02	Truck
1207	18031	V03	Truck

To table

Trip_Data

tion	Distance_km	Driver_ID	Fuel_Consum...	Origin	Trip_ID	Vehicle_ID
	1173	D01	108.42	Delhi	T001	V04
ore	1727	D08	161.33	Mumbai	T002	V06
	1459	D08	154.7	Mumbai	T003	V06

Cardinality

One to many (1:*)

Cross-filter direction

Single

☒ Make this relationship active

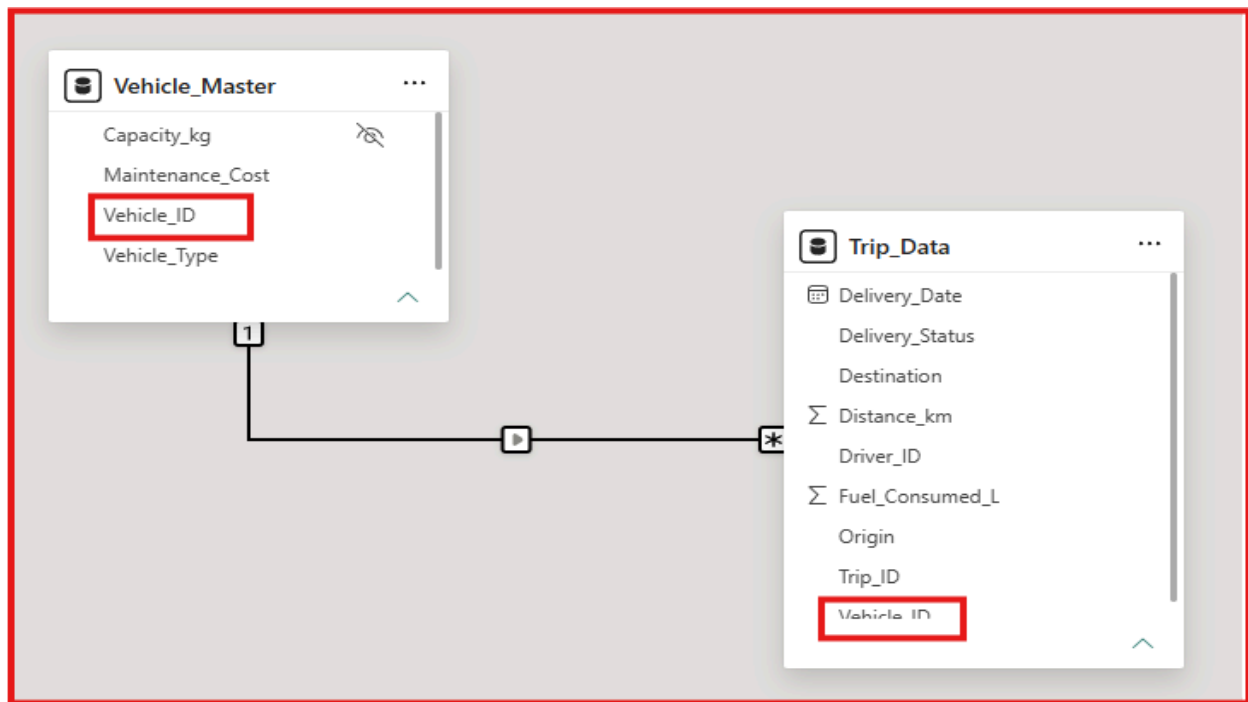
☐ Assume referential integrity

☐ Apply security filter in both directions

Save Cancel

Here the relationship shows the one to many relations between Trip_data and Vehicle master using the vehicleID column.

This allows one vehicle in the master table to be linked with multiple trip records.



Task 2. DAX Measures:

- Fuel Efficiency = Distance / Fuel Consumed

Dax measure is calculated by clicking the Table tools—>select New Measure

Structure Formatting Properties Sort Groups Relationships Calculations

Fuel_Efficiency = ([Trip_Data[Distance_km]/Trip_Data[Fuel_Consumed_L])

Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date	Fuel_Efficiency
T001	V04	D01	Delhi	Pune	1173	108.42	On-Time	27 January 2023	10.8
T002	V06	D08	Mumbai	Bangalore	1727	161.33	On-Time	21 February 2023	10.7
T003	V06	D08	Mumbai	Pune	1459	154.7	On-Time	17 February 2023	9.4
T004	V04	D09	Hyderabad	Pune	382	26.6	On-Time	18 February 2023	14.4
T005	V06	D08	Pune	Mumbai	398	33.2	On-Time	15 February 2023	12.0
T006	V06	D07	Chennai	Mumbai	1275	85.04	Late	25 February 2023	15.0
T007	V07	D03	Chennai	Kolkata	752	58.08	On-Time	19 January 2023	12.9
T008	V02	D10	Delhi	Pune	74	5.24	On-Time	01 January 2023	14.1
T009	V02	D07	Delhi	Hyderabad	186	16.22	On-Time	23 February 2023	11.5
T010	V02	D02	Bangalore	Hyderabad	1375	105.21	Late	02 February 2023	13.1
T011	V06	D03	Kolkata	Hyderabad	419	31.17	On-Time	21 January 2023	13.4
T012	V06	D01	Kolkata	Delhi	751	51.77	On-Time	15 February 2023	14.5
T013	V05	D04	Kolkata	Chennai	1571	188.52	Late	02 February 2023	8.3
T014	V05	D05	Hyderabad	Bangalore	1524	104.51	On-Time	16 February 2023	14.6
T015	V05	D06	Kolkata	Mumbai	1956	179.88	On-Time	21 January 2023	10.9
T016	V05	D06	Bangalore	Mumbai	858	91.88	Late	18 January 2023	9.3
T017	V07	D07	Pune	Kolkata	1269	102.91	On-Time	16 January 2023	12.3
T018	V07	D10	Pune	Delhi	1565	107.23	On-Time	12 February 2023	14.6
T019	V07	D10	Hyderabad	Pune	1796	155.52	On-Time	28 February 2023	11.5
T020	V04	D02	Mumbai	Bangalore	1640	148.87	On-Time	06 January 2023	11.0
T021	V07	D07	Delhi	Mumbai	446	31.98	Late	11 January 2023	13.9
T022	V07	D09	Hyderabad	Delhi	140	11.44	On-Time	07 February 2023	12.2
T023	V06	D04	Chennai	Delhi	1729	197.55	On-Time	06 February 2023	8.8

Search

- ▼ Trip_Data
 - Delivery_Date
 - Delivery_Status
 - Destination
 - Distance_km
 - Driver_ID
 - Fuel_Consumed_L
 - Fuel_Efficiency**
 - Origin
 - Trip_ID
 - Vehicle_ID
- ▼ Vehicle_Master

The formula is as follows,

Structure Formatting Properties Sort Groups Relationships Calculations

Fuel_Efficiency = ((Trip_Data[Distance_km]/Trip_Data[Fuel_Consumed_L]))

Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date	Fuel_Efficiency
T001	V04	D01	Delhi	Pune	1173	108.42	On-Time	27 January 2023	10.8
T002	V06	D08	Mumbai	Bangalore	1727	161.33	On-Time	21 February 2023	10.7
T003	V06	D08	Mumbai	Pune	1459	154.7	On-Time	17 February 2023	9.4
T004	V04	D09	Hyderabad	Pune	382	26.6	On-Time	18 February 2023	14.4
T005	V06	D08	Pune	Mumbai	398	33.2	On-Time	15 February 2023	12.0
T006	V06	D07	Chennai	Mumbai	1275	85.04	Late	25 February 2023	15.0
T007	V07	D03	Chennai	Kolkata	752	58.08	On-Time	19 January 2023	12.9
T008	V02	D10	Delhi	Pune	74	5.24	On-Time	01 January 2023	14.1
T009	V02	D07	Delhi	Hyderabad	186	16.22	On-Time	23 February 2023	11.5
T010	V02	D02	Bangalore	Hyderabad	1375	105.21	Late	02 February 2023	13.1
T011	V06	D03	Kolkata	Hyderabad	419	31.17	On-Time	21 January 2023	13.4
T012	V06	D01	Kolkata	Delhi	751	51.77	On-Time	15 February 2023	14.5
T013	V05	D04	Kolkata	Chennai	1571	188.52	Late	02 February 2023	8.3
T014	V05	D05	Hyderabad	Bangalore	1524	104.51	On-Time	16 February 2023	14.6
T015	V05	D06	Kolkata	Mumbai	1956	179.88	On-Time	21 January 2023	10.9
T016	V05	D06	Bangalore	Mumbai	858	91.88	Late	18 January 2023	9.3
T017	V07	D07	Pune	Kolkata	1269	102.91	On-Time	16 January 2023	12.3
T018	V07	D10	Pune	Delhi	1565	107.23	On-Time	12 February 2023	14.6
T019	V07	D10	Hyderabad	Pune	1796	155.52	On-Time	28 February 2023	11.5
T020	V04	D02	Mumbai	Bangalore	1640	148.87	On-Time	06 January 2023	11.0
T021	V07	D07	Delhi	Mumbai	446	31.98	Late	11 January 2023	13.9
T022	V07	D09	Hyderabad	Delhi	140	11.44	On-Time	07 February 2023	12.2
T023	V06	D04	Chennai	Delhi	1729	197.55	On-Time	06 February 2023	8.8

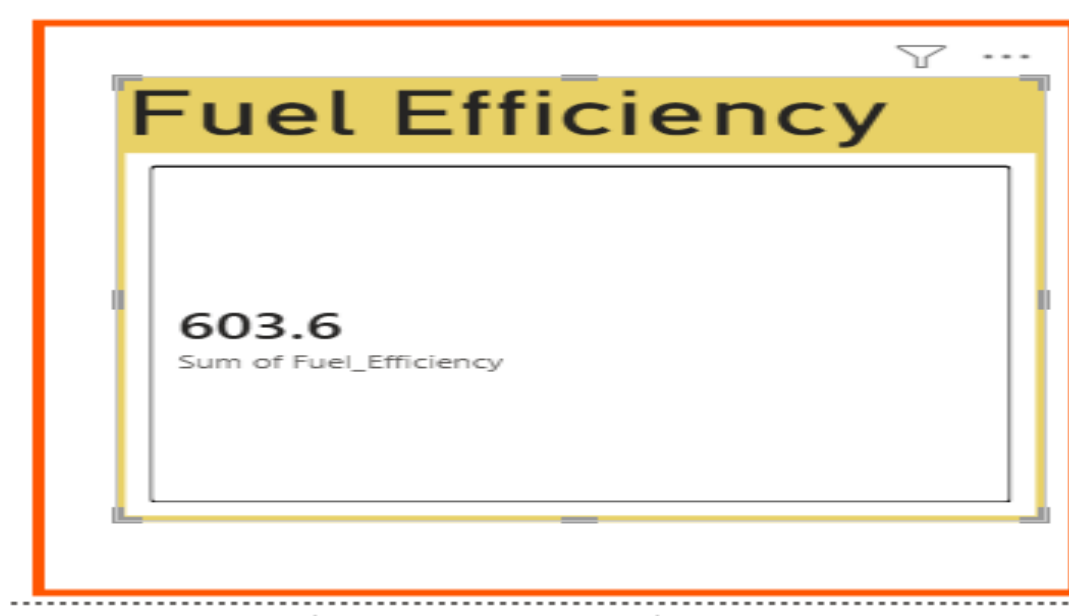
Search

- ▼ Trip_Data
 - ▼ Delivery_Date
 - Delivery_Status
 - Destination
 - Σ Distance_km
 - Driver_ID
 - Σ Fuel_Consumed_L
 - Fuel_Efficiency**
 - Origin
 - Trip_ID
 - Vehicle_ID
- ▼ Vehicle_Master

Fuel_Efficiency = (Trip_Data[Distance_km]/Trip_Data[Fuel_Consumed_L])

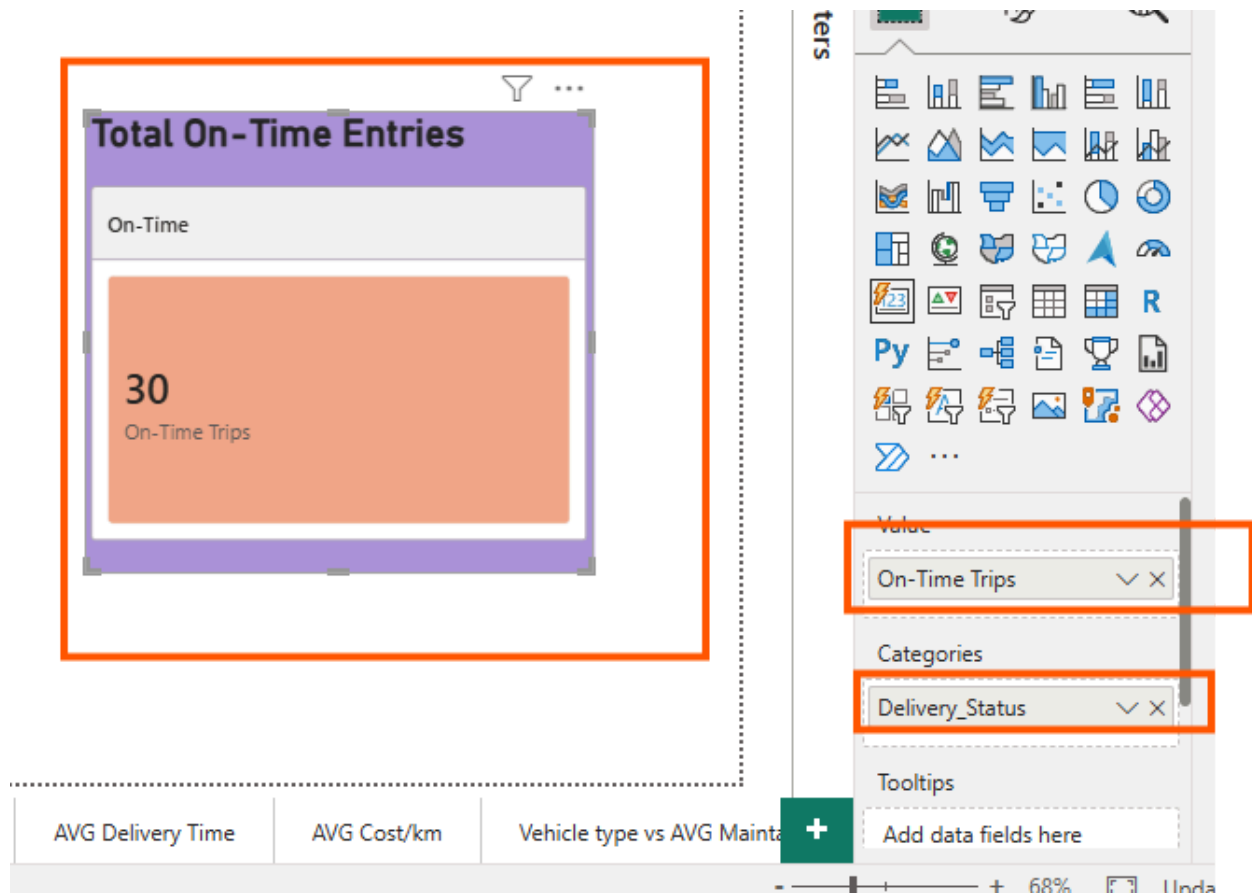
It calculates the distance travelled per litre of fuel for each trip.

In DAX calculations the measure used to get results without creating



the new column, so we visualising measure results using card view.

- **On-Time Delivery % = On-Time Trips / Total Trips:**



The Card visual is created in Power BI to display the total number of On-Time Trips. The On-Time Trips measure is used as the value, with delivery_status is used as the category. This provides a quick overview of delivery performance.

- **Cost per km = (fuel cost + Maintenance Cost) / Distance:**

The calculated column is created as cost per km using Dax function —> go to Menu bar in that click on the table tools—> select New Column

The new column is created as following formula has typed in that formula bar,

Cost per Km = `DIVIDE((([Fuel_Consumed_L] * 100) + RELATED(Vehicle_Master[Maintenance_Cost]),[Distance_km], 0)`

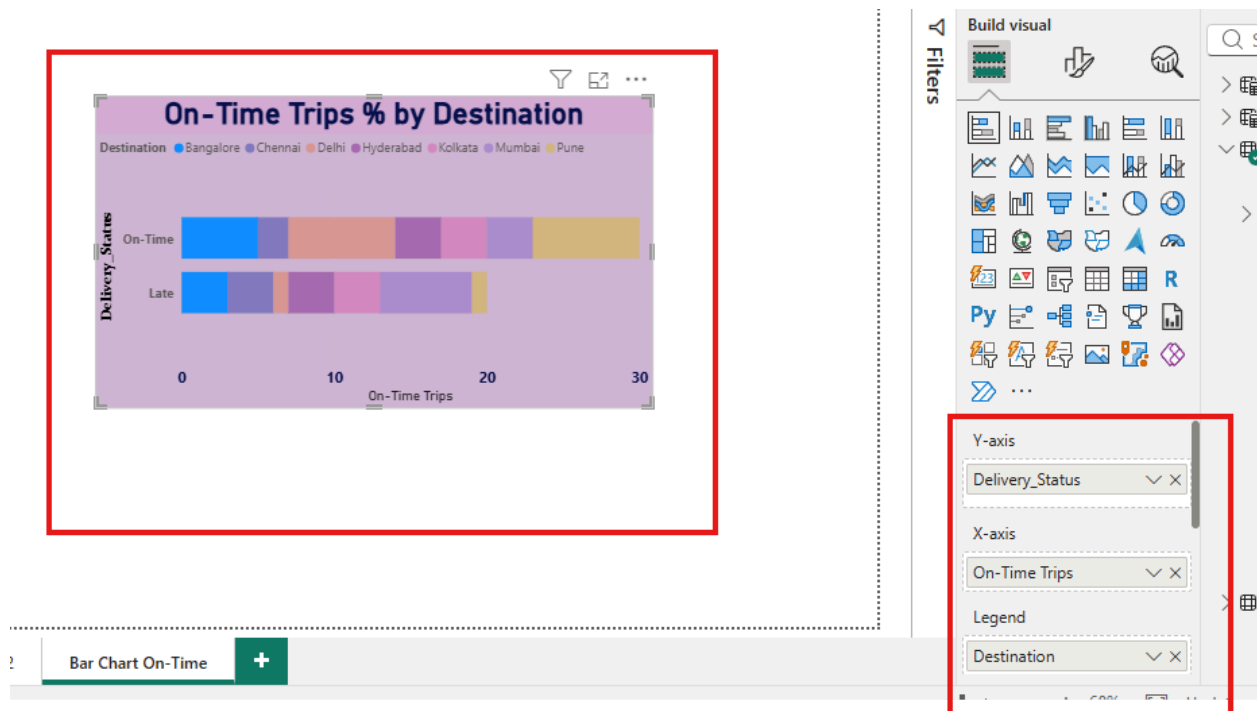
The following image shows the Cost based on the fuel consumption, maintenance cost and distance traveled.

$\text{Cost per Km} = \frac{([\text{Fuel_Consumed_L}] * 100) + \text{RELATED}(\text{Vehicle_Master}[\text{Maintenance_Cost}])}{[\text{Distance_km}]}$									
Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date	Fuel_Efficiency	Cost per Km
V04	D01	Delhi	Pune	1173	108.42	On-Time	27 January 2023	10.8	16.94
	D08	Mumbai	Bangalore	1727	161.33	On-Time	21 February 2023	10.7	12.77
	D08	Mumbai	Pune	1459	154.7	On-Time	17 February 2023	9.4	14.66
V04	D09	Hyderabad	Pune	382	26.6	On-Time	18 February 2023	14.4	30.61
V06	D08	Pune	Mumbai	398	33.2	On-Time	15 February 2023	12.0	23.20
V06	D07	Chennai	Mumbai	1275	85.04	Late	25 February 2023	15.0	11.31
V07	D03	Chennai	Kolkata	752	58.08	On-Time	19 January 2023	12.9	24.79
V02	D10	Delhi	Pune	74	5.24	On-Time	01 January 2023	14.1	98.65
V02	D07	Delhi	Hyderabad	186	16.22	On-Time	23 February 2023	11.5	45.15
V02	D02	Bangalore	Hyderabad	1375	105.21	Late	02 February 2023	13.1	12.58
V06	D03	Kolkata	Hyderabad	419	31.17	On-Time	21 January 2023	13.4	21.55
V06	D01	Kolkata	Delhi	751	51.77	On-Time	15 February 2023	14.5	14.77
V05	D04	Kolkata	Chennai	1571	188.52	Late	02 February 2023	8.3	23.30
V05	D05	Hyderabad	Bangalore	1524	104.51	On-Time	16 February 2023	14.6	18.51
V05	D06	Kolkata	Mumbai	1956	179.88	On-Time	21 January 2023	10.9	18.27
V05	D06	Bangalore	Mumbai	858	91.88	Late	18 January 2023	9.3	31.40
V07	D07	Pune	Kolkata	1269	102.91	On-Time	16 January 2023	12.3	18.23
V07	D10	Pune	Delhi	1565	107.23	On-Time	12 February 2023	14.6	15.05
V07	D10	Hyderabad	Pune	1796	155.52	On-Time	28 February 2023	11.5	15.81
V04	D02	Mumbai	Bangalore	1640	148.87	On-Time	06 January 2023	11.0	14.59
V07	D07	Mumbai	Mumbai	446	31.98	Late	11 January 2023	13.9	35.95
V07	D09	Hyderabad	Delhi	140	11.44	On-Time	07 February 2023	12.2	99.86

Task 3. Visualization:

○ Bar chart: On-Time Delivery % by Destination.

Here the visualization of the bar chart is created as a Stacked bar chart title as “On-Time Delivery % by Destination”.

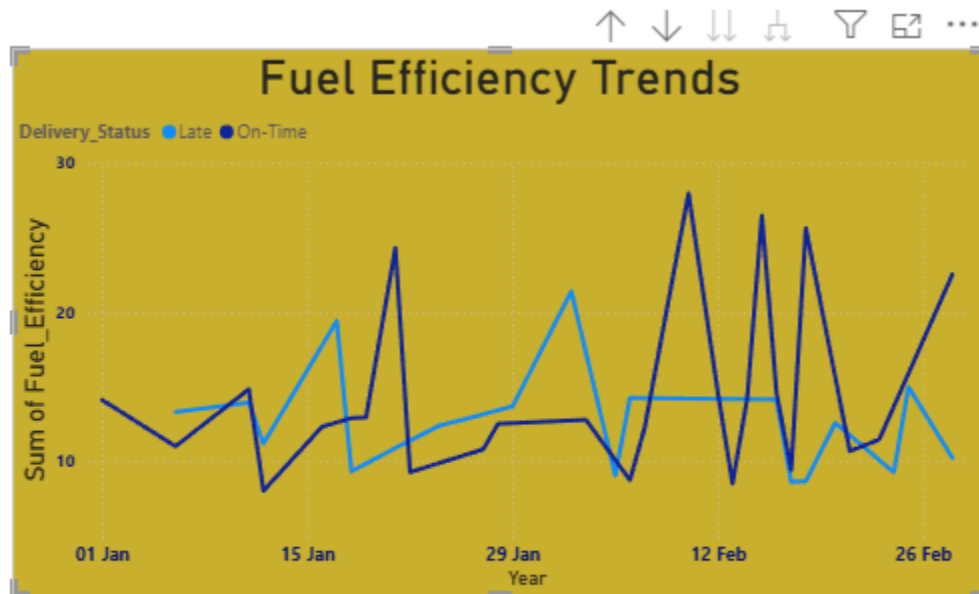


In this chart On time trip is placed in X-axis because x-axis only should be categorical data and Y-axis as Delivery status of On-time,Late.

Legend is act as a Destination to show trip distribution.This visualization helps identify which destinations have better on-time delivery performance.

○ **Line chart: Fuel efficiency trend by delivery date.**

In Power BI the line chart is used to show trends and changes in the data values.Line Chart is created to analyze how fuel efficiency changes over different dates.



In the X-axis Date,Month,Year is placed to show the timeline of January01 to February26 and the y-axis is placed as sum of Fuel_Efficiency to represent the total value of fuel.

Here the legend is Delivery_status to compare status of On time and Late.

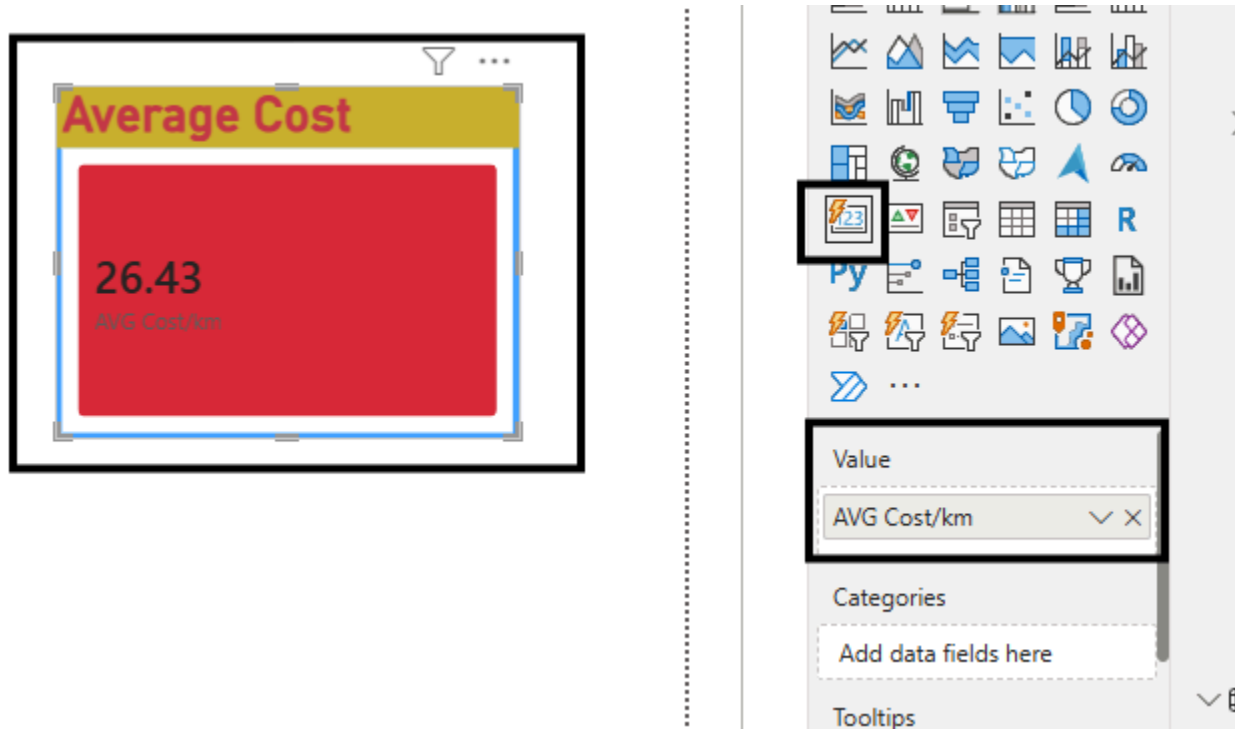
The Line shows the efficiency of the trips in the given data.This visualization is used to identify the Performance of the On Time and Late Deliveries.

○ **Cards visualization:**

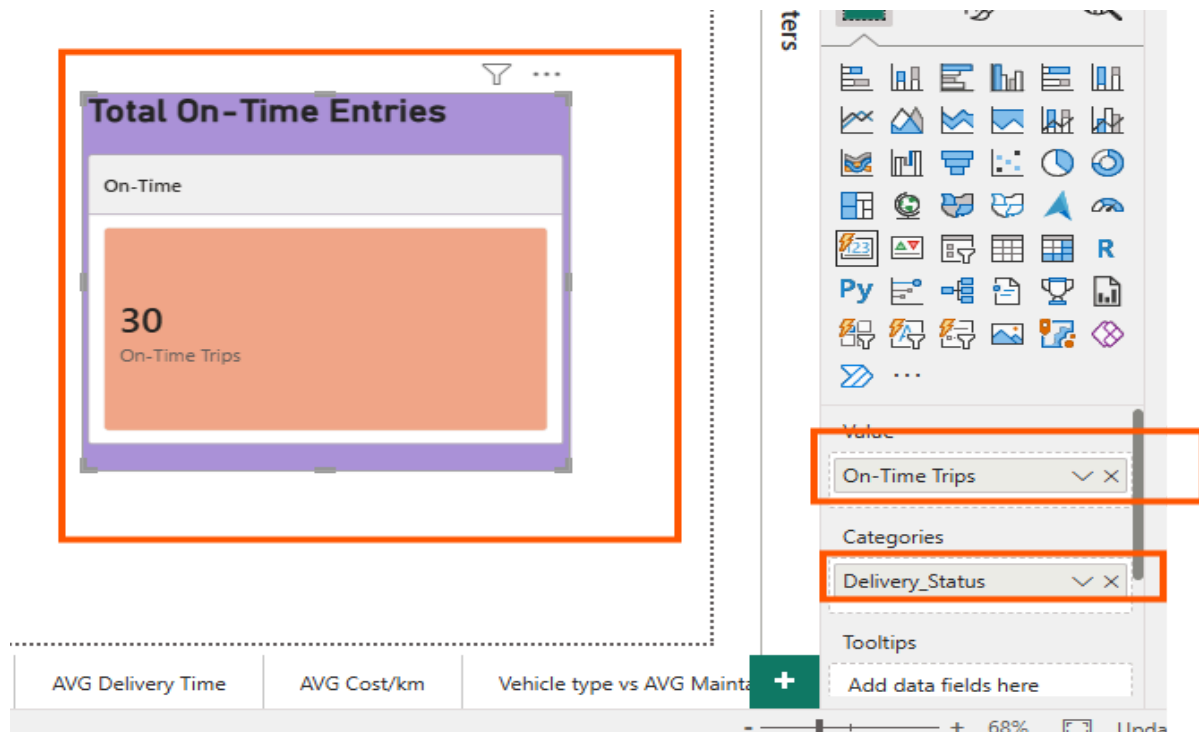
■ Average cost per km

Card visual in Power BI is used to display the Average Cost per Km.

Added the AVG/Cost measure to the **Value** field of the card.

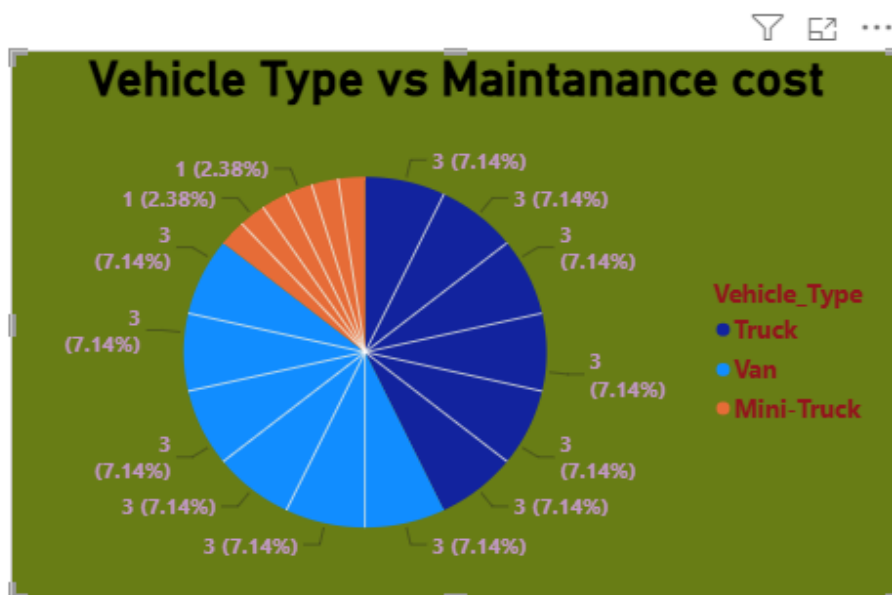


The card displays an average cost of 26.43 per km. It can help compare vehicle/trip costs and identify areas where transportation expenses are high.



○ **Pie chart: vehicle type vs Average maintenance cost:**

A pie chart in Power BI is used to represent the proportion of different categories within a total. Here the pie chart is used to analyze the contribution of Truck, Van, and Mini-Truck to the overall maintenance cost.



And also to analyze the relationship between Vehicle Type and Maintenance Cost.

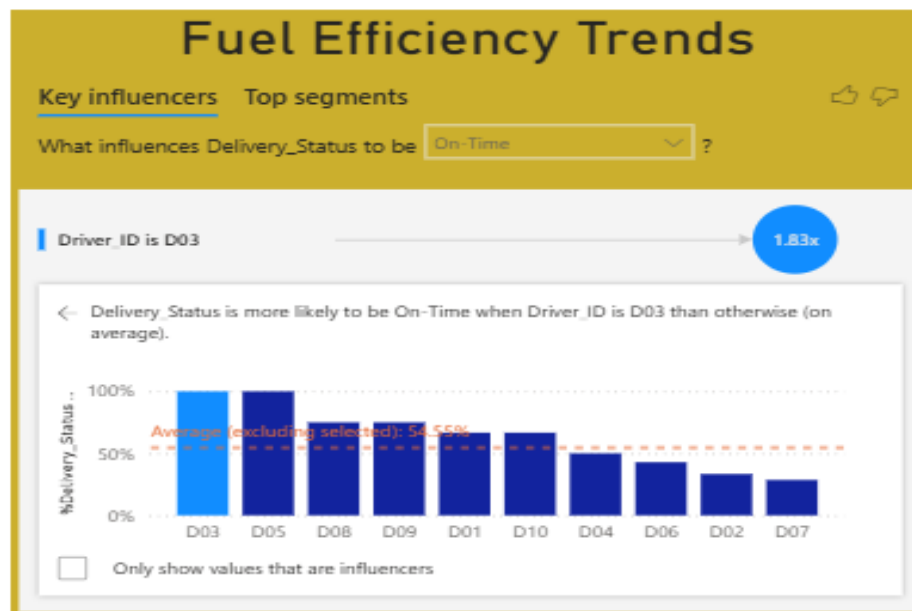
Each slice shows the maintenance-cost contribution

The percentages help to compare

the maintenance cost is distributed across the vehicle data.

Task 4. AI-Powered Visuals:

- Key Influencers Visual:
Add Delivery Status in the Analyze and explain by -distance in km, vehicle_type, Driver_ID



This is a Key Influencers visual in Power BI, used to identify factors. Here, the analysis checks what influences Delivery status.

The visual identifies Driver_ID = D03 as an important influencer.

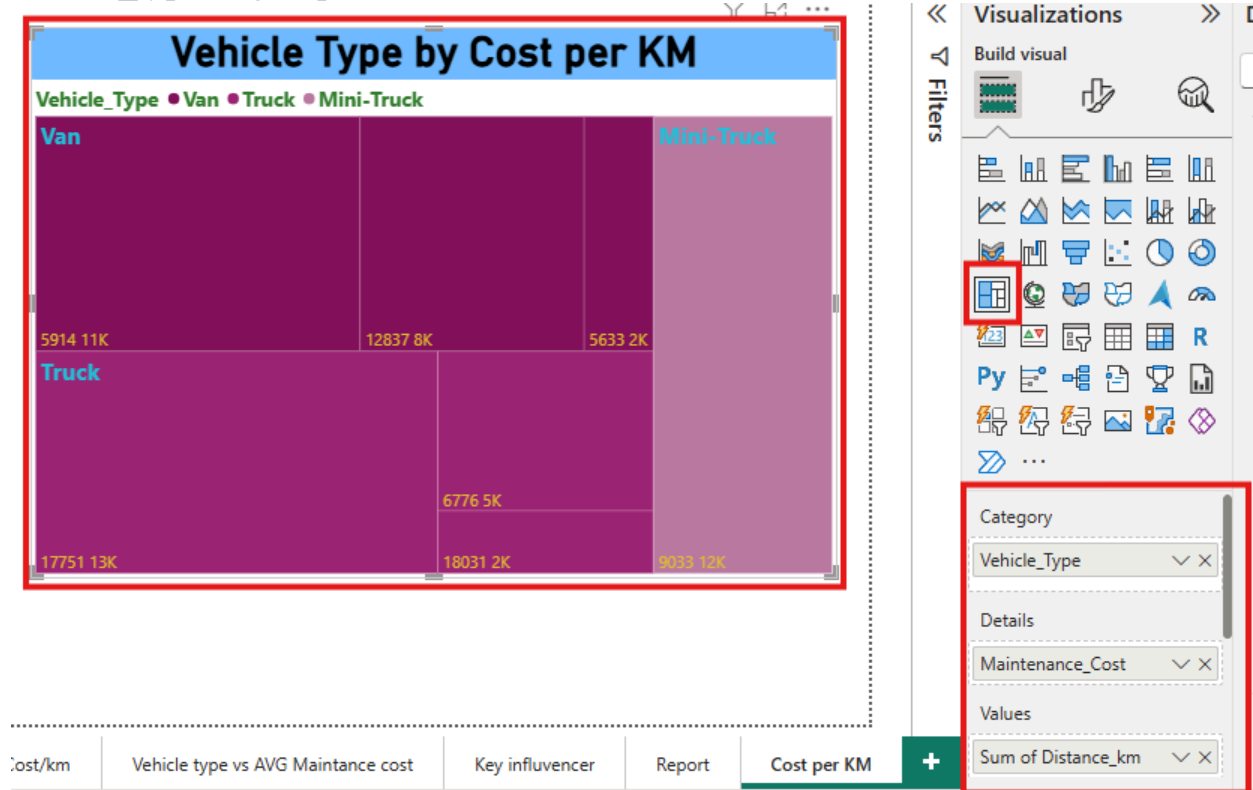
The bar chart compares different drivers based on their On-Time delivery percentage. This visual helps identify key factors affecting delivery performance and supports better operational decision-making.

❖ Decomposition Tree (AI Visual):

➤ Analyze Cost per km → explained by Vehicle Type, Maintenance_cost, and Distance_km.

Treemaps in Power BI are used for visualizing hierarchical data using nested rectangles, where size and color represent values and categories.

This is a Treemap visual in Power BI used to compare values through rectangular blocks. The category as Vehicle_type to group data into van, truck and mini truck.

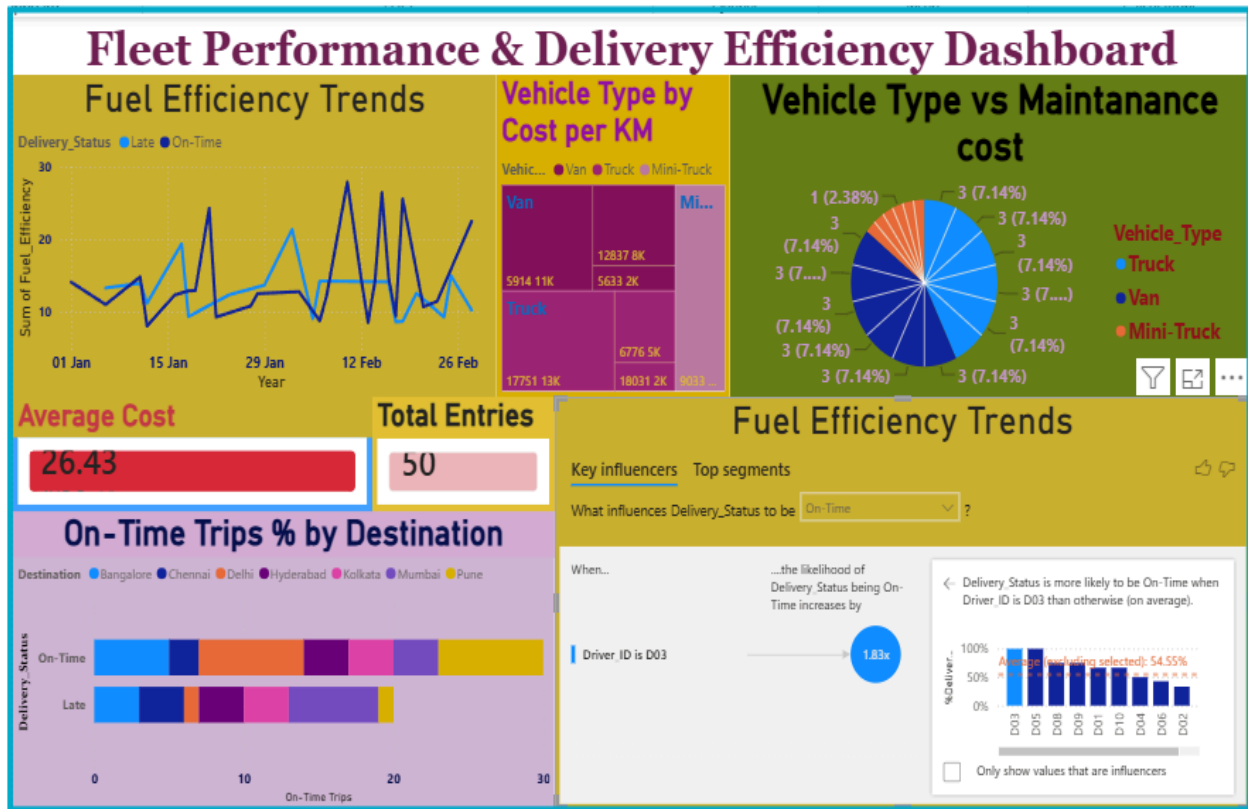


In the Details Bar the maintenance_cost provide a further breakdown within vehicle type. The values as Sum of Distance_km determine the size of each rectangle in the Treemap visual.

The different sections make it easy to compare vehicle types, maintenance costs, and distance travelled in one visual.

Report of Delivery Performance Dashboard :

This is a Fleet Performance & Delivery Efficiency Dashboard created in Power BI. It combines several visuals to analyze fuel efficiency, vehicle cost, maintenance cost, delivery performance, and factors affecting on-time delivery.



Dashboard Creation – Created interactive Power BI visuals such as line charts, treemap, pie chart, stacked bar chart, KPI cards, and Key Influencers.

Insights & Reporting – Used the dashboard to identify cost patterns, fuel-efficiency trends, delivery performance, and factors affecting on-time deliveries.